

that deviate from pre-trained knowledge, are indicative of the model’s level of adaptation to the IT dataset’s specifics. Additionally, SFT and style imitation lead the model to inaccurately include tokens from the IT dataset in its responses, negatively affecting response quality.

Finding 1. LFT and SFT learn tone imitation differently. As seen in Section 3, LFT primarily learns response initiation, and most of the response comes from pre-trained knowledge. On the other hand, SFT experiences high degrees of new token usage. This leads us to investigate if LFT learns *tone imitation* at all and the differences in *tone imitation* between models fine-tuned using the two training paradigms. Fig. 12a and 12b illustrate token distributions of common tokens at **shifted** and **marginal** positions for LLaMa-2 7B_SFT_AlpaCa 52k and LLaMa-2 7B_LFT_AlpaCa 52k. Analyzing tokens at shifted positions allows us to understand how IT affects a model to output different tokens than it would generally have without IT. As we clearly see, for LFT, shifts occur primarily in style tokens (“typically”) and response initiation tokens (“However”). On the other hand, for SFT, shifts occur in all kinds of tokens. To investigate the source of these tokens further, we perform a string search of the shifted tokens in the IT dataset. To our surprise, we found $\approx 81.2\%$ of the words that start with **shifted** tokens and 66.7% that start with **marginal** tokens borrowed from the IT dataset itself. This indicates that with SFT, models show increased borrowing of tokens from the IT dataset for response generation. We may attribute this to over-fitting on the IT dataset. A study in Section 5 further shows that these tokens are often borrowed inaccurately, leading to hallucinations.

Finding 2. Style imitation can hurt response quality. We now investigate if *style imitation* can affect response quality. We are motivated by the finding that a positive correlation exists between the length of responses in the IT dataset and the length of responses output by the fine-tuned LLM on our evaluation set (detailed results in Table 11). To achieve this, we utilize the LIMA IT dataset, which comprises responses from community Q&A forums known for their comprehensiveness, expertise, and length (refer to Table 11). When LLaMa is fine-tuned on the LIMA dataset with SFT, it produces lengthy and detailed responses, even when pre-trained knowledge might be insufficient. This often leads to hallucinations as the model strives to generate extended answers. Figure 5 showcases examples of this, with each box presenting a response from LLaMa-2 7B_SFT_LIMA 1K alongside a more concise version. We summarize our findings on style imitation: (1) As seen in Instruction 1., style imitation can sometimes enhance response quality without leading to hallucinations, particularly when the model has sufficient knowledge of the subject. (2) As seen in Instruction 2., style imitations alone can alter the nature of a response to open-ended reasoning instructions. (3) The model initially

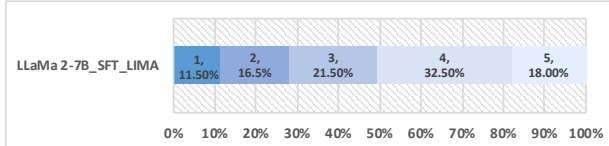
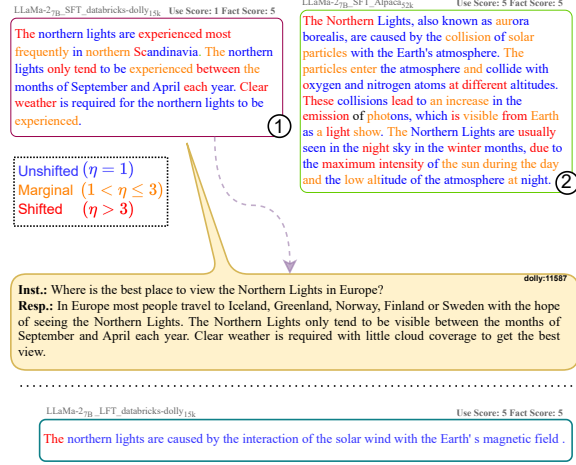


Figure 6. Human study comparing responses of a model fine-tuned on LIMA 1K and LIMA-Simple 1K. 1. Both responses are accurate, but the former is more detailed and preferred over the latter. 2. Both responses are accurate but different information-wise. 3. Both responses are completely inaccurate. 4. The former hallucinated facts in an attempt to prolong the response, while the latter did not. 5. The former hallucinated in the core facts, while the latter did not. Qualitative examples in Section B.

provides factual information but then resorts to hallucination to prolong the response or fact in the absence of adequate knowledge about the subject. These hallucinations may include randomly generated facts or content derived from the IT dataset, as seen earlier and discussed further in Section 5.

Proposed Solution: Simplifying responses in the IT dataset. One possible solution to mitigate the hallucination problem caused by style imitation is to employ LFT instead of SFT, as LFT tends to learn only stylistic elements. However, as seen in Finding 3 of Section 3, LFT doesn’t scale effectively, and SFT may act as a knowledge enhancer at larger scales (evident from LLaMa-2 7B_LFT_Tulu-V2-Mix 326k in Fig. 3). We propose an approach leveraging the strengths of both LFT and SFT. Given that LLMs possess ample pre-training knowledge for accurate response generation (Finding 4, Section 3) but struggle with comprehensive answering in pattern-copying mode, we hypothesize that SFT on IT datasets with concise but accurate responses can reduce hallucinations. We employ GPT-4 to simplify LIMA 1K, creating concise responses by removing extraneous information and term it as LIMA-Simple 1K. This model is compared with the original LLaMa-2 7B_LFT_LIMA 1K in terms of hallucination tendencies in Fig. 5. The results show a significant reduction in hallucinations beyond being concise. Quantitative results in Table 13 (rows 49-50) confirm that while the simplified model may lack depth, it surpasses the original in factuality and helpfulness. We additionally illustrate two special cases: (1) Instruction 1 illustrates a case where the model actually possessed enough knowledge to answer comprehensively and provides a more factual response with greater depth than its simplified instruction. (2) Instruction 2 demonstrates how adopting pattern copying from datasets of varied types results in divergent responses to a single open-ended reasoning-based instruction. Fig. 6 illustrates the results of a human study conducted by four expert human evaluators who manually compared the responses of LLaMa-2 7B_SFT_LIMA 1K and LLaMa-2 7B_SFT_LIMA-Simple 1K against 5 pre-defined categories. We show that simplifying responses in IT datasets for mod-

1. Instruction: What causes the northern lights?



2. Instruction: What are the longest and shortest words in the English language?

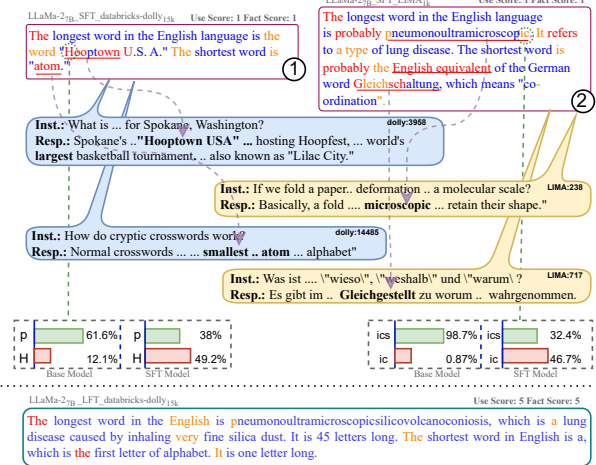


Figure 7. Illustration of hallucinations and their source of origin. We illustrate responses from LLMs fine-tuned using SFT on different IT datasets on two instances from just-eval-instruct _{1k} and show that hallucinations originate from erroneous causal links between the response and instances in the IT dataset. Models trained using SFT tend to incorrectly borrow tokens from instances in the IT dataset describing similar *concepts*. **Left.** resp. ① from LLaMa-2 _{7B} fine-tuned on databricks-dolly _{15k} is not *useful*, although *factual*, as it describes places to view the northern lights but not what causes it. The answer was directly borrowed from the IT dataset. Alpaca _{52k} has no instances related to northern lights. **Right.** Both responses have 2 factual hallucinations each, and we show that each of them originates from different instances in the IT dataset. The *shifted* and *marginal* tokens (described in Section 3) indicate that hallucinations are learned during IT. Note how models fine-tuned using LFT provide factual answers originating from pre-trained knowledge.

els significantly reduces the chances of hallucination. Quantitative results are in Table 13.

Key Takeaways: Pattern-copying increases model hallucinations. The comprehensiveness and depth of responses in the IT dataset should be phrased depending on the existing factual knowledge of the model. Additionally, pattern-copying makes the model borrow tokens from the IT dataset for responses.

5. Causal Analysis of Hallucinations

Earlier, we observed that inaccurately generated tokens in hallucinations by a model, fine-tuned using SFT, can often be traced back to the IT dataset itself. However, this raises two questions: Are such hallucinations specific to style imitation, and are inaccurate tokens borrowed random or causally driven from the IT dataset? To answer these questions, we perform a fine-grained analysis of hallucinations through the lens of causal analysis.

Finding 1. SFT increases hallucinations in a model. These hallucinations originate from erroneous causal links between the training dataset and responses. Fig. 7 illustrates 3 examples of hallucinations that do not occur in the pattern copying mode, i.e., the model hallucinates even in concise responses. Unlike the scenarios depicted in Figure 5, where hallucinations are associated with attempts to extend responses or facts, Instruction 2 in Figure 7 illustrates examples where the base pre-trained model, despite

having the correct knowledge, hallucinates after IT. Similar to prior findings, these tokens originate from the IT dataset.

Next, we establish a formal framework for a detailed analysis of such hallucinations. We ask four human experts to identify hallucinations by marking spans in the responses that are either not *factual* or *useful* in relation to the instruction. A simple string search reveals that, on average, $\approx 72\%$ of these marked phrases exist in the IT dataset, with $\approx 89\%$ of the tokens in these phrases can be classified as *shifted* or *marginal*. Next, to determine whether these hallucinatory phrases are borrowed from any IT instance by chance, we look at these hallucinations from the lens of causal analysis, considering hallucinations as the *effect* and the specific properties or attributes of the instance in the IT dataset from where the phrases were incorrectly borrowed as the *cause*.

Four expert human evaluators manually reviewed all instances in the IT dataset where the hallucinated phrases occurred. Surprisingly, they discover a pronounced propensity for the models to respond using incorrectly borrowed tokens from instances in the IT dataset that describe analogous *concepts*. For instance, response ① for Instruction 2 in Fig. 7, which seeks information about the *largest* and *shortest* words in English, partially sources its content from an unrelated instruction regarding the *longest* basketball tournament and another concerning the *smallest* atom. Similarly, response ② from the same instruction appropriates its content from an instruction discussing a related concept of "languages". Finally, response ① for Instruction 2 illustrates

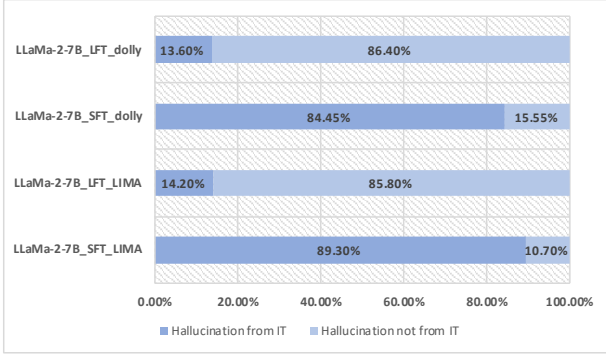


Figure 8. Human study of hallucinations by LLaMa-2 7B trained on LIMA 1K and databricks-dolly 15k and evaluated on just-eval-instruct 1k. Human evaluators found an average chance of 87% that a hallucinated phrase in a response originated from a causal relation in the IT dataset with SFT models. This phenomenon is much lesser in LFT models.

how the fine-tuned model generates a rephrased response from an instance in the IT dataset related to the same abstract concept – “about Northern lights”, but unrelated to the input instruction, but rather related to an instance in the IT dataset. We also illustrate responses of models fine-tuned with LFT and show that responses aligned to pre-trained knowledge tend to output factually correct and useful answers. Table 8 shows that expert human evaluators found an average of 87% of 1000 randomly chosen hallucinated phrases in responses by models fine-tuned using SFT to be causally related to the IT dataset. This number is only 13.9% with LFT for 500 phrases.

Finding 2. Additional findings. We further report on our findings regarding model hallucinations: (1) Detecting the causal relationships behind hallucinations proves challenging, both through manual efforts and automated methods. Sole reliance on semantic or lexical analysis often falls short in tracing the exact instance within the IT dataset that influenced the hallucinated response or its segments. (2) Quantifying the degree to which concept similarity contributes to hallucinations is difficult. Our analysis yields an average semantic similarity score of 0.418 (described in Section D) between evaluation set instances and the corresponding instances from the IT dataset, identified by human evaluators as the source of hallucinated content. Furthermore, we observed that there is only a 61.6% likelihood that a *keyword* from the hallucinated response appears in its originating instance from the IT dataset (described in Section D). Additionally, we would like to highlight the fact that post SFT-based IT, the tendency of LLMs to hallucinate with tokens from the IT dataset does not depend on the quality of the IT dataset and we observe this phenomenon throughout all IT datasets we employ in our analysis.

Qualitative Examples and Case Studies. Section B shows

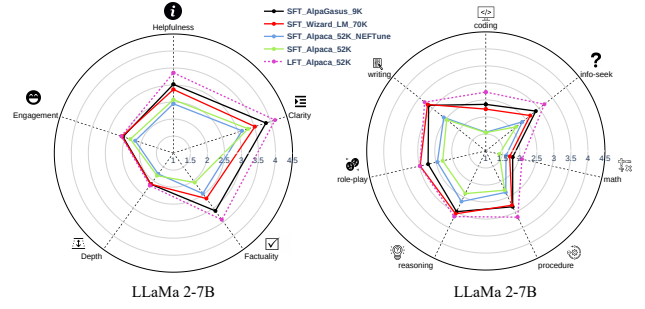


Figure 9. Comparison of various methods proposed in literature to improve IT. We show that a model trained using simple LFT outperforms all these methods across metrics and tasks.

additional examples of hallucinations, including counterfactual analysis and case studies, where we also show that models even hallucinate similarly for procedural tasks like coding and summarization.

Key Takeaways: SFT increases hallucinations in a model by making it prone to incorrectly borrow tokens from the IT dataset for responding. These tokens are borrowed from instances describing similar concepts.

6. Methods to Improve IT are Ineffective

Several enhanced IT methods have been proposed in literature for improving model response quality. In this section, we study some common methods, including AlpacaGasus (Chen et al., 2023) that employs dataset filtering, WizardLM (Xu et al., 2023a) that increases the complexity of the instructions, and NEFTune (Jain et al., 2023) that adds noise to embedding vectors while fine-tuning. Fig. 9 compares the performance of models fine-tuned with SFT using these methods (as suggested by the original papers), fine-tuned on a similar setting, with a model trained using LFT, which generates responses primarily with pre-trained knowledge. AlpacaGasus and WizardLM are improved versions of the Alpaca dataset, and for NEFTune, we also employ Alpaca. As we see, though all methods outperform SFT on Alpaca (as also suggested by the original papers), the LFT model outperforms all these models in all cases, both when averaged across individual metrics and task types. This suggests that pre-trained knowledge still dominates, and the exact benefits brought about by these methods need deeper investigation. We hypothesize that the drop in performance is mainly brought about by the increased tendency of the models to hallucinate with tokens borrowed from the IT dataset, and we urge deeper investigation by the community to solve this problem before the benefits of these tricks could enhance response quality. Beyond the scope of this section, Table 13 (Rows 45-50) also shows that LFT performance only slightly improves with these methods.

7. Related Work

While extensive research has introduced new IT datasets, models, and enhancement methods (Zhang et al., 2023a), few studies examine IT’s limitations. Concurrent work by Gudibande et al. (2023) shows that IT with datasets synthesized from powerful proprietary models only leads to imitating their style, not their knowledge. Similarly, Lin et al. (2023) shows that alignment only teaches style and proposes in-context learning as an alternative to IT that outperforms several tuned models. This can be attributed to our findings on LFT, where the models respond using pre-trained knowledge and do not lead to knowledge degradation like SFT. This supports the *superficial alignment hypothesis* by Zhou et al. (2023), positing that models gain knowledge in pre-training, with alignment shaping format used in user interactions. Finally, Kung & Peng (2023) show that models trained on simplified task definition or delusive examples achieve performance comparable to the ones trained on the original instructions. In contrast to all these works, we study the exact causes of these limitations, investigate pattern copying and hallucination from novel perspectives, and highlight the overlooked effectiveness of LFT that utilizes only pre-trained knowledge.

8. Conclusion

In this paper, we reveal various failure models of IT, including how LFT does not scale, how SFT and pattern-copying increase hallucinations, and how an LFT model outperforms various methods proposed in literature. As part of future work, we would like to propose a formal framework for detecting and mitigating hallucinations arising from SFT work on investigating novel methods of IT that can potentially improve model performance over pre-trained knowledge.

9. Limitations and Future Work

Our work has obvious limitations, including (1) Our analysis focuses solely on open-domain instruction following, and we acknowledge that fine-tuning for specific domains or tasks might enable models to gain new skills and knowledge. (2) Our analysis is limited to uni-modal language only IT, and (3) We do not study the effects of more advanced alignment methods like DPO (Rafailov et al., 2023) and RLHF and leave this for future work. (4) We do not explore retrieval-augmented generation, which decouples knowledge extraction from the model. (5) Finally, the findings of our paper are only confined to general-purpose IT datasets (datasets with an amalgamation of all kinds of general-purpose everyday instruction-response pairs) and IT datasets tailored to improve specific tasks (e.g., IT datasets tailored for imitating step-wise thinking to improve math reasoning (?). may not obey our findings.

10. Reproducibility

All LLMs evaluated in this paper were trained using LLaMA-Factory (Zheng et al., 2024b). For evaluation, we provide all prompts in Section A. We would like to emphasize that the exact replication of the numbers in our paper would require employing the prompt together with gpt-4-1106-preview, which is subject to availability by OpenAI. This is similar to a wealth of prior work that employs LLM-as-a-judge for free-form LLM response evaluation. However, if another version is employed, most trends and the core findings will still hold. Code for token-distribution analysis is directly borrowed from the repository open-sourced by Lin et al. (2023). For any questions, we request that the readers contact the corresponding author.

Impact Statement

This paper explores the limitations of Instruction Tuning for Large Language Models, which have implications that extend beyond the immediate scope of artificial intelligence research and development. By highlighting the constraints of current IT practices, our findings encourage the development of more robust and reliable conversational agents that could be employed across various sectors, including education, customer service, and accessibility technologies.

One significant impact is on the ethical use of LLMs. Our research emphasizes the importance of accuracy and factual representation in responses, reducing the risk of misinformation that can stem from hallucinations and knowledge degradation. This is particularly critical in domains where trust and veracity are paramount, such as healthcare and news dissemination.

In academia, our call for further investigation into IT’s limitations could spur a new direction of research that focuses on understanding the fundamental workings of LLMs rather than just their superficial performance improvements.

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